Analysis 1

Lecture Notes and Revision

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1 Real numbers: natural numbers and integers

Lecture date: *Add date here*

1.1 Natural numbers

For this lecture, the natural numbers start at 1:

$$\mathbb{N} = \{1, 2, 3, 4, \ldots\}.$$

Thus $0 \notin \mathbb{N}$.

Definition 1.1. A natural number n is even if $n = 2k$ for some integer k.

1.2 Operations and order on $\mathbb{N}$

The natural numbers are closed under addition and multiplication:

$$m, n \in \mathbb{N} \quad \Longrightarrow \quad m + n \in \mathbb{N} \quad \text{and} \quad mn \in \mathbb{N}.$$

For any $m, n \in \mathbb{N}$, exactly one of $m > n$, $m = n$, or $m < n$ holds. Also, $m \geq n$ means $m > n$ or $m = n$.

1.3 Properties

For $m, n, p \in \mathbb{N}$:

$$\begin{aligned} m + n = n + m, \qquad & mn = nm && \text{(commutativity)},\\ (m + n) + p = m + (n + p), \qquad & (mn)p = m(np) && \text{(associativity)},\\ m(n + p) = mn + mp \qquad & && \text{(distributivity)}. \end{aligned}$$

Order is transitive:

$$m > n \text{ and } n > p \quad \Longrightarrow \quad m > p,$$

and similarly

$$m \geq n \text{ and } n \geq p \quad \Longrightarrow \quad m \geq p.$$

1.4 Why introduce integers?

Some equations have no solution in $\mathbb{N}$. For example, $x + 1 = 0$ requires $x = -1$. This motivates the integers:

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}.$$

1.5 Questions to revisit

1.6 Revision summary